Scientific Reports

https://doi.org/10.1038/s41598-026-35964-x

Article in Press

Association between reduced chewing-induced brain blood flow and cognitive performance in mandibular prognathism patients in a pilot study

Received: 7 October 2025

Accepted: 9 January 2026

Published online: 16 January 2026

Cite this article as: Inagawa Y., Kanzaki H., Kariya C. *et al.* Association between reduced chewing-induced brain blood flow and cognitive performance in mandibular prognathism patients in a pilot study. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-35964-x

Yuri Inagawa, Hiroyuki Kanzaki, Chihiro Kariya, Saki Tanaka, Masao Kumazawa & Hiroshi Tomonari

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

Title: Association Between Reduced Chewing-Induced Brain Blood Flow and Cognitive Performance in Mandibular Prognathism Patients in a Pilot Study

Authors: Yuri Inagawa1, Hiroyuki Kanzaki1,2, Chihiro Kariya1, Saki Tanaka1, Masao Kumazawa1, and Hiroshi Tomonari1

Affiliation: 1 Department of Orthodontics, School of Dental Medicine, Tsurumi University, Yokohama 230-8501, Japan
2 Tohoku University Hospital, Department of Orthodontics, Sendai 980-8574

Corresponding author: Hiroyuki Kanzaki, Tohoku University Hospital, Department of Orthodontics, Sendai 980-8574.
E-mail: Hiroyuki.kanzaki.a8@tohoku.ac.jp

Funding: This work was supported in part by Grants-in-Aid for Scientific Research from the Japan Society for the Promotion of Science (23K09449 and 24K13186).

Data availability statement: The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

ARTICLE IN PRESS

ARTICLE IN PRESS

Abstract

Despite established links between masticatory dysfunction and cognitive impairment, cognitive function in dentofacial deformity patients remains unexplored. This study represents the first comprehensive cognitive assessment in mandibular prognathism (MP) patients.

Brain blood flow (BBF) during chewing was measured using functional near-infrared spectroscopy in MP patients versus normal occlusion controls (NORM, n=17). Cognitive function using eye-tracking technology (Mirudake) in MP patients (n=44) was compared to healthy controls (HC, n=59), with assessing six cognitive domains.

MP patients demonstrated significantly reduced chewing-induced BBF in bilateral inferior frontal gyrus compared to NORM. Global cognitive performance showed no significant difference between MP and HC groups (p=0.48). However, positive correlations existed between BBF and cognitive performance domains including global performance and memory. ROC analysis using pooled bilateral BBF data (n=88) revealed modest diagnostic potential for cognitive assessment (AUC=0.657, 95% CI: 0.431-0.865). The optimal threshold yielded 62.5% sensitivity, 73.8% specificity, and 72.7% overall accuracy for detecting cognitive impairment.

This first systematic cognitive evaluation of MP patients revealed no significant global cognitive impairment despite confirmed reductions in masticatory-induced brain activation. The observed BBF-cognition correlations suggest BBF measurements may have potential utility as an adjunct measure in cognitive screening, warranting further investigation through longitudinal studies examining orthognathic surgery's potential cognitive effects.

ARTICLE IN PRESS

ARTICLE IN PRESS

Introduction

Dentofacial deformity, characterized by significant skeletal discrepancies in the maxillofacial region, affects approximately 1-5% of the general population, with prevalence varying considerably across different ethnic groups and diagnostic criteria [1]. These skeletal abnormalities, including mandibular prognathism, retrognathism, and facial asymmetry, result not only in aesthetic concerns but also in substantial functional impairments of the stomatognathic system. Multiple studies have demonstrated that patients with dentofacial deformities exhibit significantly reduced masticatory performance compared to individuals with normal occlusion [2,3]. The masticatory efficiency in these patients has been reported to be reduced by 30-50%, with difficulties in food comminution and bolus formation [4,5].

The bidirectional relationship between masticatory function and cognitive performance is supported by converging neuroimaging, epidemiological, and interventional evidence, establishing mastication as a critical modulator of brain health [6,7]. Epidemiological studies demonstrate that tooth loss is associated with increased cognitive impairment risk in older adults, independent of socioeconomic and health factors [8]. Similarly, prospective cohort study of Japanese older adults revealed that self-reported poor dental health status was associated with increased risk of dementia onset over a 4-year follow-up period [9]. Chewing activates brain regions including sensorimotor cortex, prefrontal cortex, hippocampus, and cerebellum, with fMRI showing 15-25% increased activity [10,11]. Moreover, positron emission tomography studies demonstrate that chewing increases glucose metabolism in brain regions crucial for executive function and attention [12]. We previously reported that jaw deformity such as mandibular prognathism (MP) [13] and skeletal open bite [14] significantly attenuates chewing-induced brain blood flow (BBF) at an area reported to play an important role in cognitive function [15,16], suggesting possible retardation by malocclusion in cognitive function. Studies reveal 1.23-1.87 times higher dementia risk with poor masticatory function after adjusting for confounders [17,18]. Masticatory training improved working memory and processing speed in older adults with mild cognitive impairment [19]. Each functional tooth pair reduces dementia risk by ~3.1%, indicating a quantifiable oral-cognitive relationship [20]. These findings collectively position masticatory function not merely as a correlation but as a potentially modifiable determinant of cognitive health.

ARTICLE IN PRESS

ARTICLE IN PRESS

Despite established links between masticatory dysfunction and cognitive impairment, there remains a knowledge gap regarding cognitive function in dentofacial deformity patients, who have severe chronic masticatory impairment from developmental rather than age-related origins [21,22]. Systematic literature search reveals that while numerous studies examine functional and psychosocial impacts of dentofacial deformities, investigations of cognitive function in this population are absent, representing a significant oversight given these patients experience lifelong masticatory efficiency reductions [23]. This research gap is concerning as dentofacial deformity patients differ fundamentally from elderly populations: dysfunction occurs during critical neurodevelopmental periods, is biomechanical rather than purely dental, and affects younger individuals whose cognitive reserve and neuroplasticity may respond differently to chronic masticatory impairment.

Eye-tracking technology has emerged as a validated and sensitive biomarker for cognitive function assessment [24], with recent systematic reviews confirming its superiority over traditional screening tools [25,26]. Meta-analysis by Opwonya et al. of 35 studies showed saccadic eye movement parameters distinguished cognitive impairment with pooled sensitivity of 0.84 and specificity of 0.81, outperforming conventional cognitive screening instruments [27]. Semmelmann and Weigelt demonstrated that webcam-based eye-tracking achieves high correlation with lab equipment and sufficient resolution for detecting millisecond attention changes [28]. Validation studies show low-cost eye-tracking systems match research equipment performance, with test-retest reliability >0.85 for cognitive assessments [29]. Kaufmann et al. found eye-tracking detected cognitive deficits with 89% accuracy (vs 67% for traditional tests), reducing assessment time by 40% and eliminating ceiling effects [30]. This approach could reveal cognitive implications of masticatory dysfunction in dentofacial deformity patients, uncovering unrecognized disease burden to inform treatment strategies.

Based on these considerations, we formulated the following hypotheses for this pilot investigation. First, we hypothesized that mandibular prognathism patients would exhibit reduced chewing-induced brain blood flow in the inferior frontal gyrus compared to individuals with normal occlusion, reflecting compromised masticatory sensory input [13]. Second, we predicted that chewing-induced brain blood flow would correlate positively with cognitive performance measures, given the established role of the inferior frontal gyrus in cognitive

ARTICLE IN PRESS

ARTICLE IN PRESS

function [16]. Third, we explored whether these neurobiological alterations would be associated with detectable cognitive differences in mandibular prognathism patients compared to healthy controls, recognizing that young adults with developmental masticatory dysfunction might demonstrate different patterns than elderly populations with acquired dysfunction [31,32]. This study combined functional near-infrared spectroscopy with automated eye-tracking cognitive assessment to provide the first comprehensive examination of these hypotheses in dentofacial deformity patients. Given this knowledge gap and validated eye-tracking methodology, this study aims to evaluate cognitive function in dentofacial deformity patients using eye-tracking-based cognitive assessments compared to healthy controls. Specifically, we aim to quantify differences in multiple cognitive domains, thereby establishing the first comprehensive cognitive profile of the patient with malocclusion, specifically, MP. This investigation will not only fill a significant gap in our understanding of the systemic impacts of dentofacial deformities but also provide crucial evidence for optimizing treatment timing and developing targeted cognitive rehabilitation strategies for this patient population, potentially establishing masticatory function restoration as a modifiable factor for cognitive health maintenance across the lifespan.

ARTICLE IN PRESS

ARTICLE IN PRESS

Materials and methods

Ethical Approval

The present investigation received ethical clearance from the Institutional Review Board of Tsurumi University School of Dental Medicine under approval numbers 1316 and 121005. The study protocol adhered strictly to the ethical principles outlined in the Declaration of Helsinki regarding human research participation. Prior to enrollment, comprehensive written informed consent was secured from all study participants following detailed explanation of the research objectives, procedures, and potential risks. The study design and reporting were conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement guidelines to ensure methodological rigor and transparency in human observational research.

Sample Size Calculation

Sample size calculation was conducted prior to data collection using the statistical power analysis software, G*Power (version 3.1.9.4) (Heinrich-Heine-Universität Düsseldorf, Germany, https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower). The analysis was based on an anticipated effect size of 0.8, with alpha set at 0.05 and statistical power at 0.8. Based on our previous study [13], which demonstrated significant differences in chewing-induced brain blood flow between mandibular prognathism patients and normal occlusion subjects ($p < 0.05$ in the inferior frontal gyrus), we estimated an effect size of 0.8 for sample size calculation. This effect size is supported by the consistently large effects observed in occlusal function parameters (occlusal contacts: $p < 0.01$; masseter EMG: $p < 0.05$) in our prior work. This yielded a minimum required sample size of 54 participants for sum of two groups, which matched our final sample size of (61 for BBF comparison, 103 for cognitive function comparison) participants.

Subjects

The experimental group consisted with the patients with MP (MP group, n = 44), comprising 21 men and 23 women (age: 25.0 ± 10.4, range: 15-59 years). The MP group comprised patients who presented to our hospital for initial consultation regarding surgical orthodontic treatment. Participants were

ARTICLE IN PRESS

ARTICLE IN PRESS

required to meet the following inclusion criteria:

-Absence of congenital abnormalities

-Complete permanent dentition excluding third molars

-Presence of anterior crossbite

-Skeletal and dental midline deviation ≤1.0 mm from the facial midline

-Absence of functional symptoms, including temporomandibular joint disorders

-No previous history of orthodontic treatment

-Angle Class III molar relationship with mandibular first molar positioned anteriorly relative to the maxillary first molar

-Skeletal Class III intermaxillary relationship as determined by ANB angle

Exclusion criteria included systemic diseases or concurrent medication use that could potentially influence cerebrovascular regulation or neuromuscular function.

All participants completed a comprehensive health questionnaire and underwent a detailed clinical interview to verify adherence to the specified inclusion and exclusion criteria prior to enrollment. The health questionnaire assessed the following domains: (1) current systemic diseases under treatment (including cardiovascular, cerebrovascular, neurological, endocrine, and psychiatric disorders); (2) medical history of conditions that could affect cerebrovascular regulation or neuromuscular function; (3) current medications, particularly those influencing central nervous system function, blood pressure, or vascular tone; (4) history of head trauma or neurological disorders; and (5) temporomandibular joint symptoms or dysfunction. Participants reporting any systemic diseases or medication use were excluded from the study.

ARTICLE IN PRESS

Control group for BBF comparison

A control group (NORM group) comprising 17 participants (4 males, 13 females; mean age: 19.4 ± 0.6 years, range: 19-21 years) was established based on the following inclusion criteria:

-Complete dentition excluding third molars (no missing teeth other than wisdom teeth)

-Normal overjet and overbite measurements (2-4 mm for each parameter, indicating appropriate occlusion in the anterior region)

-Skeletal and dental midline deviation ≤1.0 mm from the facial midline (absence of significant lateral jaw deviation)

ARTICLE IN PRESS

-Absence of functional disorders, including temporomandibular joint dysfunction
-No prior history of orthodontic intervention
-Angle Class I molar relationship (indicating favorable anteroposterior relationship between maxillary and mandibular first molars)
-Normal intermaxillary relationships, as evidenced by cephalometric measurements (ANB angle: 3.1 ± 2.5□; mandibular plane to Frankfort horizontal plane angle: 26.6 ± 6.7□)

Control group for cognitive function comparison

Data from 59 healthy participants (29 males, 30 females, age: 44.8 □ 3.0; range: 40-49 years) were provided by Ai-BrainScience Inc. as healthy control (HC) group data (unpublished data). These data were collected under the approval of the Ethics Committee of Yoga Allergy Clinic (approval number: AI11001XX6). Prior to enrollment, comprehensive written informed consent was also obtained from all study participants following detailed explanation of the research objectives, procedures, and potential risks.

The following inclusion criteria were applied for control group selection:
-Age 40 years or older and younger than 50 years at the time of consent acquisition
-No history or comorbidity of dementia, active neurological diseases, or psychiatric disorders at the time of consent acquisition

Participants meeting any of the following criteria were excluded from the study:
-Individuals with ophthalmological conditions that significantly impair eye-tracking detection (e.g., strabismus, cataracts resulting in insufficient visual acuity, or visual field defects due to glaucoma or diabetes), rendering eye-tracking detection impossible. Participants using the following vision correction devices were required to perform testing without these devices: 1) Sunglasses with insufficient brightness or difficulty in color recognition. 2) Eyeglass frames that are extremely small or excessively thick. 3) High myopia (-6.25 diopters or greater) with significant refractive correction lenses that impede eye-tracking. 4) Post-cataract surgery correction lenses (+13 diopters approximately) with significant refractive correction that impedes eye-tracking. 5) Bifocal or multifocal lenses with convex lenses for near vision in the lower portion.
-Neurological disorders with motor impairments that clearly compromise the ability to maintain posture during measurement

ARTICLE IN PRESS

ARTICLE IN PRESS

-Individuals who had undergone the measurement of cognitive function by Mirudake within 3 months prior to this study
-Other conditions including nystagmus or trauma that prevent appropriate measurement of gaze data, or individuals unable to maintain posture during testing

BBF Measurement

BBF was assessed using functional near-infrared spectroscopy (fNIRS; OEG-16 device; Spectratech, Tokyo, Japan), which quantifies oxygenated hemoglobin (oxy-Hb) concentrations in the prefrontal cortex through 16 channels. Detailed methodological procedures for BBF measurements have been previously documented [13,14]. Based on a priori hypotheses, we focused our primary analysis on the bilateral inferior frontal gyrus (IFG) channels (channels 1-4 for right IFG and channels 13-16 for left IFG) for the following reasons. First, our previous investigation demonstrated that chewing-induced brain blood flow increases were most prominent in the IFG compared to other cortical regions [13]. Second, the IFG has been consistently implicated in cognitive function, particularly executive function and cognitive control processes [16]. Given that the primary objective of this study was to investigate whether dentofacial deformities affect cognitive function through alterations in mastication-induced brain activation, the IFG represented the most theoretically relevant region of interest. While all 16 channels were recorded during measurements, statistical analyses for group comparisons and correlation analyses with cognitive performance focused on the bilateral IFG channels to test our specific hypotheses regarding the mastication-cognition relationship in mandibular prognathism patients.

Prior to experimental procedures, measurement calibration was performed and demonstrated high reproducibility of BBF responses during chewing tasks, with intraclass correlation coefficients (ICC) exceeding 0.7 after a one-week interval [13]. To ensure consistent fNIRS data acquisition across all experimental trials, standardized probe placement protocols were implemented using conventional headgear systems. Participants received explicit instructions to maintain stable head positioning throughout the measurement period to minimize motion-related artifacts. All measurements were conducted in a controlled, quiet environment with reduced external stimuli to eliminate potential confounding variables.

ARTICLE IN PRESS

ARTICLE IN PRESS

BBF were measured in NORM and MP groups.

Task Setting for fNIRS

A simple block design was employed, consisting of baseline rest periods and experimental conditions. The positive control condition comprised a cognitive arithmetic task, wherein participants performed verbal serial subtraction of 7 from 100. The experimental condition involved masticating soft CRT paraffin (Ivoclar Vivadent, Tokyo, Japan) using their preferred masticatory side. Each condition was executed for approximately 60 seconds, with participants maintaining their natural masticatory rhythm, and the peak Oxy-Hb value obtained during this period was used as the Oxy-Hb value for each task in the comparative analysis. To ensure adequate consistency in chewing patterns, all sessions were monitored by the investigator, and participants received instructions to utilize their preferred masticatory side consistently across all trials. Although individual masticatory frequencies were not rigidly standardized, participants were instructed to sustain a comfortable and consistent rhythm throughout each condition to reduce intra-individual measurement variability. Following each experimental condition, a minimum 30-second rest period was implemented to permit oxy-Hb concentrations to return to baseline levels. Baseline recovery was verified through continuous oxy-Hb signal monitoring prior to initiating subsequent conditions. Representative temporal change of BBF was shown in Figure 1A.

ARTICLE IN PRESS

fNIRS Data Analyses

Comprehensive methodological details for fNIRS data analysis have been previously documented [13,14]. Baseline correction and moving averages were calculated to eliminate artifacts. In summary, peak oxy-Hb values within each channel during respective experimental conditions constituted the primary analytical focus of this investigation. Peak oxy-Hb values were computed for each condition and subsequently subjected to inter-group comparative analysis and correlation analysis against cognitive function domains. For quantification of peak oxy-Hb concentrations, the median value derived from the maximum concentrations recorded across channels 1-4 and channels 13-16 was designated as the representative metric for each participant.

Measurement of cognitive function (Mirudake measurement).

ARTICLE IN PRESS

We measured the cognitive function of the subjects with an eye-tracking application, Mirudake (Ai-BrainScience Inc., Osaka, Japan). Briefly, Mirudake is an eye-tracking application designed for iPadOS 12.1 or later that utilizes ARKit-based technology for gaze measurement. We used an iPad Pro (12.9-inch, 264 ppi) mounted inverted on a stand with the front camera capturing participants' eye movements. The device was positioned 51 cm from participants at a 90-degree angle to the table.

The measurement protocol begins with face position adjustment, where participants align themselves with an on-screen guide before raw gaze data collection commences. To ensure accurate mapping of gaze data to screen coordinates, Mirudake employs a five-point calibration system followed by an accuracy verification phase. To maintain sufficient precision for cognitive assessment, the system requires that the Euclidean distance between on-screen targets and estimated gaze positions falls within predetermined thresholds during accuracy verification. This study implemented thresholds of either 200 points (38 mm) or 300 points (57 mm), with prior validation confirming no significant differences in cognitive function scores between these settings. When participants fail the accuracy check, the system automatically reinitiated the five-point calibration sequence. This calibration-verification cycle continues until either successful completion or three consecutive failures, at which point the application displays a warning and returns to the initial face position adjustment screen. Participants who failed to achieve acceptable accuracy after four complete cycles (comprising up to 12 calibration and verification attempts) were excluded from the study. The HC and MP group underwent cognitive assessment using the identical Mirudake system and protocol. All assessments were conducted in the same laboratory setting by trained examiners following standardized procedures. Testing was performed individually in a quiet, well-lit room, with participants receiving standardized instructions and practice trials prior to formal assessment to ensure task comprehension.

The assessment evaluated six primary cognitive domains: memory, executive function, attention, visuospatial cognition, language, and orientation. Global cognitive performance score was derived by averaging performance scores from the six assessed cognitive domains. Mirudake is the validated digital cognitive assessment tool certified by Japan's Ministry of Health, Labour and Welfare as medical software (certification number: 30500BZX00235000).

ARTICLE IN PRESS

ARTICLE IN PRESS

The system has demonstrated moderate to good test-retest reliability with an intraclass correlation coefficient of 0.74 in previous validation studies [24]. For clinical interpretation and comparison with established cognitive screening tools, Mirudake scores were converted to MMSE-equivalent scores using the validated conversion formula provided in the official product data sheet available from the Pharmaceuticals and Medical Devices Agency (PMDA), Japan. The conversion was performed using the regression equation: $y = 2.3x - 10.3$, where x represents the MMSE score and y represents the Mirudake score. This conversion formula was derived from validation studies comparing Mirudake performance with standard MMSE assessments in Japanese populations.

Cognitive functions were measured in MP group, and the data were compared to HC data provided by Ai-BrainScience Inc.

Statistical Analyses

Statistical computations described below were executed using SPSS Statistics software (version 27.0, IBM, Tokyo, Japan). Multiple statistical approaches were employed to ensure robustness of findings across different analytical frameworks. Mann-Whitney U test was selected for group comparisons due to its non-parametric nature, avoiding normality assumptions and minimizing the influence of potential outliers in brain blood flow measurements. Two one-sided tests (TOST) for equivalence testing were specifically employed to provide positive evidence for group equivalence in cognitive function, as traditional null hypothesis testing cannot demonstrate equivalence and non-significant p-values do not prove absence of differences. For correlation analyses, both Spearman rank correlation (capturing monotonic relationships, robust to outliers) and Pearson correlation with bootstrap resampling (1000 iterations, assessing linear relationships with robust confidence intervals) were performed to comprehensively characterize relationships between variables. Convergence of results across multiple methods was considered evidence for robustness of findings. The distribution of sex (male/female) across the three groups was compared using Pearson's chi-square test.

Comparison between groups

For between-group comparisons, we utilized a dual analytical approach combining traditional non-parametric methods (Mann–Whitney U test) with

ARTICLE IN PRESS

ARTICLE IN PRESS

bootstrap resampling techniques to ensure comprehensive and reliable interpretation of results. Given the non-normal distribution characteristics of our dataset and the need for outlier-resistant analysis, the Mann–Whitney U test served as our primary comparative method. To enhance the robustness of our statistical inferences, particularly considering our moderate sample size, we supplemented this approach with bootstrap analysis involving 1000 resampling iterations. This dual methodology provided both conventional hypothesis testing (with statistical significance set at $p < 0.05$) and detailed assessment of effect reliability through bootstrap-derived confidence intervals (CI). We considered between-group differences statistically meaningful when the 95% CI for the median difference excluded zero, demonstrating consistent directional effects.

Equivalence test

Equivalence testing was performed with the setting of equivalence margins using distribution-based methodology [33]. Two margins were established: 1. conservative margin: ±0.2 □ pooled standard deviation (SD), 2. standard margin: ±0.5 □ pooled SD. Equivalence was assessed using TOST procedure, which tests whether the group difference falls within the predetermined equivalence margins. The TOST procedure consists of:

1. Lower bound test: Tests whether the HC group mean is not substantially lower than the MP group mean
2. Upper bound test: Tests whether the HC group mean is not substantially higher than the MP group mean

Statistical equivalence was declared if both one-sided tests yielded p-values < 0.05. Additionally, the 90% CI for the group difference was examined. Equivalence was supported if the entire interval fell within the equivalence margins.

Correlation analysis

For examining relationships between variables, we conducted correlation analyses using both Spearman's rank correlation and Pearson correlation methods, each enhanced with bootstrap resampling (1000 iterations) to improve estimate stability.

Evaluation of the diagnostic performance of BBF measurements in cognitive assessment

ARTICLE IN PRESS

ARTICLE IN PRESS

Evaluation of the diagnostic performance of BBF measurements in cognitive assessment were performed as follows.

1. To maximize statistical power and account for potential bilateral variations, data from both left and right sides were pooled, resulting in a total sample size of 88 measurements (n = 88) for the subsequent analyses.
2. The threshold value for classification was established using cognitive function measurements obtained through the Mirudake assessment tool. The Mirudake comprehensive scores were converted to Mini-Mental State Examination (MMSE) equivalent values using established conversion tables [34]. The threshold was set as the global cognitive performance score 50, which corresponding to the lower limit of the normal range in MMSE (i.e., MMSE ≥ 26), representing the boundary between normal cognitive function and cognitive impairment.
3. Receiver operating characteristic (ROC) curve analysis was performed to evaluate the diagnostic performance of BBF measurements in discriminating between cases below and above the predetermined threshold. The area under the ROC curve (AUC) was calculated as the primary measure of discriminative ability.
4. CIs for the AUC were calculated using the bootstrap method with 1000 resampling iterations. This non-parametric approach was selected due to the relatively small number of positive cases (n = 4), as bootstrap methods provide more robust CI estimates compared to normal approximation methods in small sample studies. Bootstrap resampling was performed with replacement from the original dataset, and the 95% CI was determined using the 2.5th and 97.5th percentiles of the bootstrap distribution.
5. The optimal threshold for diagnostic classification was determined using Youden's index (sensitivity + specificity - 1), which maximizes the sum of sensitivity and specificity. Additional performance metrics, including sensitivity, specificity, positive predictive value, negative predictive value, and overall accuracy, were calculated at the optimal threshold.

ARTICLE IN PRESS

ARTICLE IN PRESS

Results

Baseline demographic characteristics.

Regarding the sex ratio, no statistically significant differences were observed between groups (MP: 21 male and 23 female, NORM: 4 male and 13 female, HC: 29 male and 30 female, Pearson's chi-square test: $\chi^2 = 3.71$, df = 2, P = 0.1565). However, significant differences in mean age were found among the three groups (NORM (19.4 ± 0.6, range: 19-21), MP (25.0 ± 10.4, range: 15-59), and HC (44.8 ± 3.0, range: 40-49); $p < 0.05$). Therefore, prior to the following analysis, we examined whether age influences BBF and cognitive function within each group. BBF exhibited no correlation with age in either group. Cognitive function also showed no correlation with age in both the HC and MP group. These findings indicating no correlation between age and either BBF or cognitive function suggest that despite the observed age differences between groups, the influence of age is limited. Consequently, the subsequent between-group comparisons can be considered to reflect differences arising from the distinct characteristics of each group rather than age-related confounding effects.

Although comprehensive masticatory function testing could not be systematically conducted for all participants due to equipment limitations, electromyographic (EMG) data for masseter muscle activity during maximum voluntary clenching were successfully obtained from a subset of MP patients (n=24 out of 44). The mean EMG amplitude for the available MP subset was 165.0 ± 6.0 μV, which was statistically comparable to our previously published MP cohort (159.8 ± 34.8 μV, n=25, p = 0.65) [13] and substantially lower than the normal occlusion reference values (266.1 ± 30.0 μV, n=11) reported in that study.

Mandibular prognathism attenuates chewing-induced BBF.

To examine the effects of mandibular prognathism on chewing-induced BBF, we measured regional BBF in the bilateral inferior frontal gyrus during paraffin chewing tasks (Figure 1B). Statistical analysis revealed significant differences in chewing-induced BBF between NORM and MP in both hemispheres. In the right inferior frontal gyrus, NORM demonstrated significantly higher percent induction of BBF compared to MP (median [IQR]: NORM 57% [20-80%] vs. MP 23% [10-45%]; $p < 0.05$). Similarly, in the left inferior frontal gyrus, NORM showed significantly greater activation than MP (median [IQR]: NORM 50% [15-75%] vs. MP 33% [5-60%]; $p < 0.05$).

ARTICLE IN PRESS

ARTICLE IN PRESS

Both Mann-Whitney U test and bootstrap analysis confirmed the statistical significance of these bilateral differences, demonstrating robust evidence for attenuated chewing-induced BBF in patients with MP. This observed pattern was consistent across both hemispheres, suggesting a generalized reduction in cortical activation during mastication in MP.

These findings replicate our previous report [13] and provide further evidence that MP significantly impacts the neural mechanisms underlying masticatory function, as reflected by reduced BBF responses in the inferior frontal gyrus during chewing activities.

Comparison of cognitive function scores between HC and MP.

To assess whether MP gives any influence on cognitive function, we compared Mirudake scores in both HC and MP. Statistical analysis revealed no significant difference in global cognitive performance score ($p = 0.4776$, 95% CI: [-2.8, 5.9], Figure 2A). The median of global cognitive performance scores was comparable between groups, with HC showing a median score of approximately 73 and MP showing a median score of approximately 70. Despite the lack of statistical significance, descriptive analysis revealed a numerical tendency for lower global cognitive performance score in MP compared to HC.

To further clarify whether there is any difference between HC and MP in global cognitive performance score, equivalence analysis was performed (Table 1). Conservative margin did not demonstrate statistical equivalence (lower bound $p = 0.032$, upper bound $p = 0.377$) in the TOST procedure. The 90% CI exceeded the conservative equivalence margin (±2.2 points), as the upper bound (5.15) was greater than +2.2. On the other hand, standard margin (±5.5 points) confirmed statistical equivalence (lower bound $p = 0.001$, upper bound $p = 0.035$). The entire 90% CI fell within the standard equivalence margins. This result suggests that while MP may exhibit a subtle tendency toward reduced cognitive function as measured by Mirudake scores, this difference does not reach statistical significance and may not represent a clinically meaningful impairment in cognitive function assessment.

We further compared each of the six primary cognitive domains (Figures 2B to 2G). Almost all domains exhibited no statistical difference between the groups. However, orientation score shows higher value in MP as compared to that in HC, with statistical significance (Figure 2G).

ARTICLE IN PRESS

ARTICLE IN PRESS

Correlation analysis between BBF and cognitive functions.

We then performed correlation analysis between BBF and cognitive functions (Table 2). BBF at both the right and the left inferior frontal gyrus exhibited some positive correlations to cognitive function scores, such as global cognitive performance and memory. Furthermore, visuospatial cognition score indicates positive correlation to BBF at the right side, though executive function score indicates positive correlation to BBF at the left side. Higher BBF values were associated with higher cognitive performance scores, while lower BBF values were associated with lower cognitive performance scores.

Correlation analysis among cognitive function parameters within MP.

Correlation analyses between cognitive function parameters were conducted using both Spearman and Pearson with bootstrap resampling methods. Both analytical approaches demonstrated similar overall patterns of associations among the cognitive domains (Figure 3). The strongest correlation was observed between the global cognitive performance score and memory function (Spearman r = 0.74, Pearson r = 0.75). The global cognitive performance score also showed moderate correlations with executive function (Spearman r = 0.62, Pearson r = 0.66) and language (Spearman r = 0.49, Pearson r = 0.58). Among the individual cognitive domains, memory demonstrated a moderate correlation with language (Spearman r = 0.55, Pearson r = 0.46). Executive function showed weak to moderate correlations with orientation in both analyses (Spearman r = 0.39, Pearson r = 0.36).

Notably, correlations involving visuospatial cognition showed discrepancies between the two analytical methods, with several correlations reaching significance in the Pearson analysis (e.g., visuospatial cognition-global cognitive performance score: r = 0.61) but not in the Spearman analysis (r = 0.40). Many inter-domain correlations, particularly those involving attention, visuospatial cognition, and orientation, did not reach statistical significance in either analysis.

Evaluation of the diagnostic performance of BBF measurements in cognitive assessment.

ROC curve analysis was performed to evaluate the discriminative ability of BBF measurements for cognitive status classification (Figure 4). The area under the ROC curve (AUC) was 0.657 (95% CI: 0.431-0.865, bootstrap method

ARTICLE IN PRESS

ARTICLE IN PRESS

with 1000 resampling iterations). This AUC value indicates discriminative ability above chance level (AUC = 0.5), suggesting that BBF measurements may have potential utility as an adjunct measure in cognitive screening. Despite the lack of statistical significance, the effect size analysis revealed a moderate effect (Cohen's d = 0.510, 95% CI: -0.221 to 1.240), suggesting clinically meaningful differences between groups. The wide CI reflects the uncertainty inherent in small sample analyses, with the interval encompassing both null and substantial effects. Power analysis indicated that approximately 61 subjects per group would be required to detect the observed effect size with 80% statistical power at α = 0.05, compared to the current 8 subjects in the below-threshold group. The estimated statistical power of the current study was approximately 82%, suggesting adequate power despite the non-significant result.

Threshold performance analysis revealed multiple viable cutoff values (Table 3), with Youden's index ranging from 0.109 to 0.363, indicating varying trade-offs between sensitivity and specificity across different threshold selections.

The optimal diagnostic threshold was determined using Youden's index, which maximizes the sum of sensitivity and specificity (Table 4). The optimal BBF threshold was identified as ≤0.068174, yielding the following performance characteristics. Sensitivity: 62.5% (95% CI: 29.0-96.0%), Specificity: 73.8% (95% CI: 64.1-83.4%), Positive predictive value: 19.2% (95% CI: 4.1-34.4%), Negative predictive value: 95.2% (95% CI: 89.8-100.0%), Overall accuracy: 72.7% (95% CI: 63.4-82.0%), and Area Under Curve: 0.657 (95% CI: 0.431-0.865).

ARTICLE IN PRESS

ARTICLE IN PRESS

Discussion

This study represents the first comprehensive investigation of cognitive function in dentofacial deformity patients, yielding three principal findings that address critical knowledge gaps regarding systemic impacts of severe masticatory dysfunction. First, MP patients did not exhibit statistically significant cognitive impairment compared to healthy controls. Equivalence analysis revealed that groups existed at the threshold between equivalence and non-equivalence, with conclusions dependent on margin stringency. Using conservative criteria (±0.2 SD), equivalence was not demonstrated, whereas standard criteria (±0.5 SD) supported equivalence. The observed mean difference of 1.56 points, while statistically non-significant ($p = 0.555$), represents 71% of the conservative margin and 28% of the standard margin, placing groups in an intermediate zone. The negligible effect size (Cohen's $d = 0.142$) supports that any between-group difference lacks practical significance, and these borderline results emphasize the critical importance of a priori equivalence margin specification and suggest that larger sample sizes may provide more definitive evidence. Second, and significantly, chewing-induced brain blood flow was reduced in MP patients' bilateral inferior frontal gyrus regions despite the absence of clinically detectable cognitive impairment at the group level. Third, correlation analyses revealed positive associations between BBF and multiple cognitive domains, including global performance and memory, establishing a neurobiological link between masticatory-induced cerebral activation and cognitive function. These findings provide the first empirical evidence for altered neurocognitive processing patterns in dentofacial deformity patients.

Regarding the potential clinical utility of BBF measurements, this study demonstrates that they exhibit modest diagnostic performance for cognitive assessment, with an area under the curve of 0.657 (95% CI: 0.431-0.865). According to established criteria, AUC values are interpreted as: 0.90-1.0 (excellent), 0.80-0.89 (good), 0.70-0.79 (fair), 0.60-0.69 (poor), and 0.50-0.59 (fail) [35]. Our AUC of 0.657 places BBF in the "poor" category, yet above the threshold for potential clinical utility. Many clinically useful biomarkers demonstrate AUC values in the 0.6-0.8 range, particularly in complex conditions where multiple factors contribute to outcomes [36]. The primary constraint is the small number of below-threshold cases (n=8), resulting in wide confidence intervals. Power analysis indicates approximately 60 subjects per group would

ARTICLE IN PRESS

ARTICLE IN PRESS

be required for adequate statistical power. Despite these limitations, the moderate effect size (Cohen's d = 0.510) suggests clinically meaningful differences between groups, supporting BBF's biological plausibility as a cognitive biomarker. While BBF alone may not serve as a standalone diagnostic test, it shows potential value as a screening tool or component of multimodal approaches. Threshold analysis revealed multiple viable cutoff options (Youden's index: 0.109-0.363), providing flexibility for clinical implementation. The MP subjects were relatively young without many pathological subjects with cognitive decline. Therefore, we investigated detection of subjects with values below Mirudake conversion values, corresponding to MMSE lower normal limits. Future studies require large-scale populations including participants with cognitive impairments. This pilot study establishes preliminary evidence for BBF's diagnostic potential in cognitive assessment. While the modest AUC value and sample size limitations prevent definitive clinical recommendations, the above-chance discriminative ability, moderate effect size, and methodological rigor justify continued investigation in larger validation studies.

These findings should be interpreted within the established research literature linking masticatory dysfunction to cognitive impairment, while recognizing important distinctions from previous investigations. Dintica et al.'s 22-year longitudinal study of 544 adults demonstrated that poor masticatory ability accelerated spatial and fluid cognitive decline after age 65 [31]. Similarly, Sta Maria et al.'s systematic review and meta-analysis of studies from 2000-2022 confirmed that older adults with masticatory dysfunction exhibited increased risk of cognitive impairment, with consistent evidence across multiple cognitive domains [32]. However, research investigating mastication-cognition relationships in younger adult populations, especially patients with malocclusion, remains notably limited, with most studies focusing on acute effects of gum chewing [37] rather than chronic masticatory dysfunction. The few young adult studies, such as Kim et al.'s investigation of chewing effects on brain antioxidants and cognition, examined temporary enhancement from masticatory stimulation rather than chronic impairment consequences [38]. Importantly, Sta Maria et al. found that gum chewing reduced cognitive processing speed in young adults, suggesting age-dependent mastication-cognition relationships [32]. The present investigation represents the first systematic cognitive evaluation of dentofacial deformity patients, who experience developmental rather than age-related masticatory dysfunction.

ARTICLE IN PRESS

ARTICLE IN PRESS

This distinction is crucial, as dentofacial deformity patients demonstrate biomechanical masticatory impairment [39], potentially producing different cognitive impacts compared to elderly populations with acquired masticatory dysfunction. The absence of significant cognitive differences in our young adult MP population, despite demonstrable BBF reductions, may reflect age-related differences in neuroplasticity and cognitive reserve that warrant further investigation.

Several mechanistic pathways may explain the observed reduction in chewing-induced brain blood flow in mandibular prognathism patients. The primary mechanism likely involves impaired stomatognathic function and reduced masticatory sensory input. Our previous investigations demonstrated that occlusal contact area and jaw movement parameters significantly correlate with masticatory-induced BBF responses [13,14]. These findings suggest that the compromised occlusal function characteristic of dentofacial deformities directly impacts neural activation patterns during mastication, as evidenced by reduced periodontal mechanoreceptor stimulation and diminished masticatory muscle activity [13,14]. Given that mastication induces trigeminal sensory input [40], the mechanistic pathway likely involves malocclusion-induced reduction in effective masticatory stimulation leading to decreased trigeminal sensory input, subsequently attenuating ascending arousal signals that normally enhance prefrontal cortical blood flow during chewing. While our findings demonstrate significant correlations between BBF measures and cognitive performance domains, suggesting a neurobiological link between masticatory dysfunction and cognitive processing, the cross-sectional design precludes definitive causal inferences. The hypothetical mechanism whereby reduced masticatory stimulation leads to decreased BBF, potentially impacting cognitive function, represents a plausible neurobiological pathway warranting investigation through longitudinal and interventional studies to establish temporal relationships and causal directionality. Beyond these acute sensory pathways, chronic masticatory dysfunction may induce more fundamental neuroplastic changes through sensory deprivation mechanisms. The face primary motor cortex (FACE-M1) and primary somatosensory cortex (FACE-S1) demonstrate remarkable plasticity in response to altered oral sensory input [41]. Animal studies have revealed that tooth loss or occlusal disruption triggers reorganization of cortical sensorimotor representations, with neuroimaging evidence showing widespread volumetric changes in regions processing somatosensory, motor,

ARTICLE IN PRESS

ARTICLE IN PRESS

and cognitive functions [42,43]. These cortical reorganizations may involve both adaptive and maladaptive plasticity: while some compensatory changes may initially preserve function, prolonged sensory deprivation can lead to shrinkage of cortical representation areas and altered connectivity patterns [41]. At the molecular level, reduced masticatory stimulation has been associated with downregulation of brain-derived neurotrophic factor (BDNF) signaling pathways and suppressed hippocampal neurogenesis in animal models [44], suggesting that chronic masticatory dysfunction may compromise the neurobiological substrates essential for learning and memory consolidation. Notably, some evidence suggests these neuroplastic changes may be partially reversible with restoration of proper occlusal function [42], though complete recovery to pre-dysfunction states may not occur. In the context of developmental dentofacial deformities such as MP, where occlusal dysfunction begins during critical periods of brain maturation, these neuroplastic adaptations may result in distinct cortical organization patterns compared to acquired masticatory dysfunction in adulthood, potentially explaining the absence of overt cognitive impairment despite measurable BBF reductions in our young adult population.

Interestingly, MP patients showed significantly higher orientation scores than controls, contrasting with reduced performance trends in other cognitive domains. This counterintuitive result may reflect compensatory enhancement of spatial-temporal awareness in response to altered oral sensory feedback and warrants careful mechanistic consideration. Several potential explanations exist for this unexpected finding. First, since trigeminal sensory pathways connect to the parietal cortex, which has reciprocal hippocampal connections for spatial processing [45,46], chronic trigeminal alterations might indirectly influence spatial cognitive abilities. Second, MP's developmental nature during critical brain maturation periods [47,48] may facilitate compensatory reorganization that enhances orientation functions while compromising others. Third, eye-tracking orientation assessment may capture different processes than traditional cognitive tasks [49], potentially reflecting enhanced visual-spatial strategies compensating for altered orofacial proprioception, as MP patients' experience with facial asymmetry may enhance spatial orientation performance. However, this interpretation requires caution as it may reflect measurement artifacts or sample-specific characteristics. Further investigation with diverse paradigms is needed, given the absence of similar findings in previous research.

This study has several limitations that warrant consideration and careful

ARTICLE IN PRESS

ARTICLE IN PRESS

interpretation of findings. First, because of the cross-sectional nature of this pilot study, the observed associations should not be interpreted as evidence of causality, and further exploration is mandatory to clarify causality. Whether BBF reductions preceded, followed, or developed concurrently with cognitive changes remains unclear, limiting causative mechanism inference and preventing capture of developmental trajectories or adaptive changes in MP patients. Second, modest sample sizes (MP: n=44; NORM: n=17; HC: n=59) constrained statistical power in multiple ways. Post-hoc power analysis revealed insufficient power (9.1%) to detect the observed cognitive effect size, increasing Type II error risk. While no significant difference was found between HC and MP groups, this null finding cannot be interpreted as demonstrating equivalence given the low statistical power. To achieve 80% power for detecting small-to-medium effects (Cohen's d = 0.3), approximately 176 participants per group would be required. Alternatively, if the observed negligible effect represents the true population difference, results suggest groups may be practically equivalent, warranting formal equivalence testing with appropriate sample sizes. Additionally, as a pilot study with relatively small sample size, the ROC analysis for BBF diagnostic performance yielded wide confidence intervals (95% CI: 0.431-0.865), substantially limiting the generalizability and precision of our findings. The moderate AUC value (0.657) indicates that chewing-induced brain blood flow alone may not be sufficient as a standalone screening tool for cognitive assessment. However, considering (1) the non-invasive nature of fNIRS, (2) our previous findings showing significant BBF attenuation in MP patients [13], and (3) established associations between masticatory function and cognitive health in aging populations, this approach deserves further exploration in adequately powered studies. Future research should examine whether combining multiple parameters (e.g., BBF, eye-tracking metrics, and occlusal function measures) might improve predictive accuracy for cognitive screening purposes. Third, this pilot study did not apply multiple comparison corrections (e.g., false discovery rate or Bonferroni correction) to the statistical analyses. Given the exploratory nature of this pilot investigation and the moderate sample size, we prioritized minimizing Type II error (false negatives) to avoid missing potentially important relationships in this first systematic assessment of cognitive function in dentofacial deformity patients. However, this approach increases the risk of Type I error (false positives), and findings should be interpreted with this caveat in mind. Notably, our primary findings—

ARTICLE IN PRESS

ARTICLE IN PRESS

reduced BBF in MP patients and correlations between BBF and global cognitive performance/memory—demonstrated consistent significance across multiple analytical methods (Mann-Whitney U test and bootstrap analysis for group comparisons; Spearman and Pearson correlations for associations), supporting their robustness. Nevertheless, future confirmatory studies with larger sample sizes should employ appropriate corrections for multiple comparisons to rigorously validate these preliminary findings. Fourth, demographic differences between groups present important confounding factors. Most notably, the HC group was significantly older (mean: 44.8 ± 3.0 years; range: 40-49 years) than the MP group (mean: 25.0 ± 10.4 years; range: 15-59 years; $p < 0.001$). This age discrepancy arose because the HC database was originally collected for establishing normative values in cognitive screening, with a minimum age threshold of 40 years, whereas MP patients undergoing orthognathic surgery are predominantly in their 20s to 40s. Critically, given well-established age-related cognitive decline, this age difference likely renders our findings conservative rather than inflated: older HC participants may exhibit lower cognitive performance than younger age-matched controls would demonstrate, potentially masking cognitive differences that might be detected with appropriate age-matching. Indeed, examination of the HC database reveals declining cognitive performance trends across successive age decades (40s, 50s, 60s, 70s). Therefore, if age-matched controls in their 20s-30s had been available for comparison, statistically significant cognitive impairment in MP patients might have been detected. The current null finding should thus be interpreted cautiously, as it represents comparison against an older, potentially lower-performing control group rather than truly age-matched peers. Additionally, education level data were unavailable for the both groups due to database limitations, precluding assessment of this important demographic variable's potential influence. Selection bias from clinical MP patient recruitment, which may have resulted in sampling of more severe cases seeking surgical intervention rather than the full spectrum of MP severity, represents another potential confounder. Furthermore, the eye-tracking cognitive assessment approach, while validated in other populations, has limited validation specifically against traditional neuropsychological batteries in dentofacial deformity populations [49]. Future confirmatory studies require age-matched control groups with comprehensive demographic data collection, including education level, socioeconomic status, and other relevant covariates,

ARTICLE IN PRESS

ARTICLE IN PRESS

to provide definitive evidence regarding cognitive function in dentofacial deformity populations. Fifth, comprehensive masticatory function assessment, including electromyography and bite force measurements, was not systematically conducted across all participants due to equipment replacement during the study period. While partial EMG data were successfully collected from a subset of MP patients and demonstrated weak masseter muscle activity consistent with our previous findings in MP patients and significantly lower than normal occlusion controls [13], the absence of complete functional data for all participants, particularly the control groups, limits direct verification of masticatory impairment in the current cohort. However, several factors support the presence of masticatory dysfunction in our MP group: (1) clinical diagnosis of skeletal Class III malocclusion requiring orthognathic surgery inherently indicates compromised occlusal function; (2) well-documented evidence in the literature demonstrating reduced occlusal contacts, bite force, and muscle activity in MP patients [50-52]; (3) observed malocclusion severity (ANB angle: -2.7 ± 2.5°, anterior crossbite); and (4) consistency between available partial data and previous systematic measurements. Nevertheless, future studies should include comprehensive, standardized masticatory function assessment across all participant groups to provide more robust verification of functional impairment. Finally, unmeasured variables including socioeconomic status beyond the HC group, appearance-related anxiety, sleep quality, and other lifestyle factors may have influenced cognitive performance without systematic control, highlighting the need for more comprehensive covariate assessment in future investigations.

Despite these limitations, several critical research directions emerge from these findings that could substantially advance understanding of the mastication-cognition relationship in dentofacial deformity populations. While no significant cognitive impairments were found in MP patients, the observed correlations between BBF and cognitive performance suggest that the potential relationship between orthognathic surgery and cognitive enhancement warrants investigation, as benefits may extend beyond aesthetic and functional outcomes. However, rigorous validation through intervention studies is required before clinical implications can be established. Specifically, future research should prioritize: (1) longitudinal studies comparing cognitive function before and after orthognathic surgery to establish temporal relationships and potential causality; (2) large-scale cross-sectional investigations with enhanced statistical

ARTICLE IN PRESS

ARTICLE IN PRESS

power to detect subtle effects and confirm current findings; (3) mechanistic neuroimaging studies elucidating precise neural pathways between masticatory dysfunction and cognitive processing; (4) randomized controlled trials examining orthognathic surgery's cognitive effects with appropriate control groups; and (5) intervention studies exploring non-surgical masticatory rehabilitation approaches as potential cognitive interventions. These research directions would investigate orthognathic surgery's potential cognitive effects while providing the validation necessary for clinical translation through prospective intervention research with appropriate controls.

In conclusion, this pilot study provides the first exploratory assessment of cognitive function in mandibular prognathism patients. Despite acknowledged statistical limitations including modest sample sizes and resulting wide confidence intervals, our findings suggest that MP patients do not exhibit marked global cognitive impairment, although reductions in masticatory-induced brain blood flow were observed. The correlations between BBF and cognitive performance, while preliminary, suggest potential neurobiological relationships worthy of further investigation. However, given the cross-sectional design and statistical uncertainties inherent in this pilot investigation, these findings should be interpreted cautiously and do not establish causal relationships. Rather, they provide preliminary empirical evidence that warrants validation through larger, adequately powered longitudinal studies examining whether orthognathic surgery influences cognitive outcomes. Understanding the broader systemic implications of dentofacial deformities beyond aesthetic and functional considerations remains an important area for continued investigation in oral and maxillofacial medicine, with potential implications for comprehensive patient care and treatment planning pending confirmation through rigorous future research.

ARTICLE IN PRESS

ARTICLE IN PRESS

Acknowledgements

The authors thank Ai-BrainScience Inc. (Osaka, Japan) for generously providing cognitive function data from healthy control participants. This work was supported in part by Grants-in-Aid for Scientific Research from the Japan Society for the Promotion of Science (23K09449 and 24K13186).

ARTICLE IN PRESS

ARTICLE IN PRESS

References

1 Proffit, W. R., Fields, H. W., Jr. & Moray, L. J. Prevalence of malocclusion and orthodontic treatment need in the United States: estimates from the NHANES III survey. *The International journal of adult orthodontics and orthognathic surgery* **13**, 97-106 (1998).

2 van den Braber, W., van der Glas, H., van der Bilt, A. & Bosman, F. Masticatory function in retrognathic patients, before and after mandibular advancement surgery. *Journal of oral and maxillofacial surgery : official journal of the American Association of Oral and Maxillofacial Surgeons* **62**, 549-554, doi:10.1016/j.joms.2003.06.016 (2004).

3 Iwase, M., Ohashi, M., Tachibana, H., Toyoshima, T. & Nagumo, M. Bite force, occlusal contact area and masticatory efficiency before and after orthognathic surgical correction of mandibular prognathism. *International journal of oral and maxillofacial surgery* **35**, 1102-1107, doi:10.1016/j.ijom.2006.08.014 (2006).

4 Kobayashi, T., Honma, K., Shingaki, S. & Nakajima, T. Changes in masticatory function after orthognathic treatment in patients with mandibular prognathism. *The British journal of oral & maxillofacial surgery* **39**, 260-265, doi:10.1054/bjom.2000.0576 (2001).

5 Nakata, Y. *et al.* Changes in stomatognathic function induced by orthognathic surgery in patients with mandibular prognathism. *Journal of oral and maxillofacial surgery : official journal of the American Association of Oral and Maxillofacial Surgeons* **65**, 444-451, doi:10.1016/j.joms.2005.12.071 (2007).

6 Weijenberg, R. A., Lobbezoo, F., Knol, D. L., Tomassen, J. & Scherder, E. J. Increased masticatory activity and quality of life in elderly persons with dementia--a longitudinal matched cluster randomized single-blind multicenter intervention study. *BMC Neurol* **13**, 26, doi:10.1186/1471-2377-13-26 (2013).

7 Kossioni, A. E. The Association of Poor Oral Health Parameters with Malnutrition in Older Adults: A Review Considering the Potential Implications for Cognitive Impairment. *Nutrients* **10**, doi:10.3390/nu10111709 (2018).

8 Lexomboon, D., Trulsson, M., Wårdh, I. & Parker, M. G. Chewing ability and tooth loss: association with cognitive impairment in an elderly population study. *J Am Geriatr Soc* **60**, 1951-1956, doi:10.1111/j.1532-5415.2012.04154.x (2012).

ARTICLE IN PRESS

ARTICLE IN PRESS

9 Yamamoto, T. *et al.* Association between self-reported dental health status and onset of dementia: a 4-year prospective cohort study of older Japanese adults from the Aichi Gerontological Evaluation Study (AGES) Project. *Psychosomatic medicine* **74**, 241-248, doi:10.1097/PSY.0b013e318246dffb (2012).

10 Luraschi, J. *et al.* Neuroplasticity in the adaptation to prosthodontic treatment. *J Orofac Pain* **27**, 206-216, doi:10.11607/jop.1097 (2013).

11 Sesay, M., Tanaka, A., Ueno, Y., Lecaroz, P. & De Beaufort, D. G. Assessment of regional cerebral blood flow by xenon-enhanced computed tomography during mastication in humans. *Keio J Med* **49 Suppl 1**, A125-128 (2000).

12 Momose, T. *et al.* Effect of mastication on regional cerebral blood flow in humans examined by positron-emission tomography with ^{15}O-labelled water and magnetic resonance imaging. *Archives of oral biology* **42**, 57-61, doi:10.1016/s0003-9969(96)00081-7 (1997).

13 Kanzaki, H. *et al.* Mandibular prognathism attenuates brain blood flow induced by chewing. *Sci Rep* **9**, 19104, doi:10.1038/s41598-019-55553-5 (2019).

14 Kariya, C. *et al.* Skeletal Anterior Open Bite Attenuates the Chewing-Related Increase in Brain Blood Flow. *Dentistry Journal* **12**, 161 (2024).

15 Maillet, D. & Rajah, M. N. Association between prefrontal activity and volume change in prefrontal and medial temporal lobes in aging and dementia: a review. *Ageing research reviews* **12**, 479-489, doi:10.1016/j.arr.2012.11.001 (2013).

16 Nyberg, L. Cognitive control in the prefrontal cortex: A central or distributed executive? *Scandinavian journal of psychology* **59**, 62-65, doi:10.1111/sjop.12409 (2018).

17 Galindo-Moreno, P. *et al.* The impact of tooth loss on cognitive function. *Clinical oral investigations* **26**, 3493-3500, doi:10.1007/s00784-021-04318-4 (2022).

18 Saito, S. *et al.* Association between tooth loss and cognitive impairment in community-dwelling older Japanese adults: a 4-year prospective cohort study from the Ohasama study. *BMC Oral Health* **18**, 142, doi:10.1186/s12903-018-0602-7 (2018).

19 Okamoto, N. *et al.* Association between tooth loss and the development of mild memory impairment in the elderly: the Fujiwara-kyo Study. *J Alzheimers Dis* **44**, 777-786, doi:10.3233/jad-141665 (2015).

ARTICLE IN PRESS

ARTICLE IN PRESS

20 Kiuchi, S. *et al.* Oral Status and Dementia Onset: Mediation of Nutritional and Social Factors. *Journal of dental research* **101**, 420-427, doi:10.1177/00220345211049399 (2022).

21 Pachêco-Pereira, C., Pereira, J. R., Dick, B. D., Perez, A. & Flores-Mir, C. Factors associated with patient and parent satisfaction after orthodontic treatment: a systematic review. *American journal of orthodontics and dentofacial orthopedics : official publication of the American Association of Orthodontists, its constituent societies, and the American Board of Orthodontics* **148**, 652-659, doi:10.1016/j.ajodo.2015.04.039 (2015).

22 Soh, C. L. & Narayanan, V. Quality of life assessment in patients with dentofacial deformity undergoing orthognathic surgery--a systematic review. *International journal of oral and maxillofacial surgery* **42**, 974-980, doi:10.1016/j.ijom.2013.03.023 (2013).

23 Abrahamsson, C., Henrikson, T., Bondemark, L. & Ekberg, E. Masticatory function in patients with dentofacial deformities before and after orthognathic treatment-a prospective, longitudinal, and controlled study. *European journal of orthodontics* **37**, 67-72, doi:10.1093/ejo/cju011 (2015).

24 Oyama, A. *et al.* Novel Method for Rapid Assessment of Cognitive Impairment Using High-Performance Eye-Tracking Technology. *Sci Rep* **9**, 12932, doi:10.1038/s41598-019-49275-x (2019).

25 Wolf, A., Tripanpitak, K., Umeda, S. & Otake-Matsuura, M. Eye-tracking paradigms for the assessment of mild cognitive impairment: a systematic review. *Front Psychol* **14**, 1197567, doi:10.3389/fpsyg.2023.1197567 (2023).

26 Stuart, S. *et al.* The Measurement of Eye Movements in Mild Traumatic Brain Injury: A Structured Review of an Emerging Area. *Front Sports Act Living* **2**, 5, doi:10.3389/fspor.2020.00005 (2020).

27 Opwonya, J. *et al.* Saccadic Eye Movement in Mild Cognitive Impairment and Alzheimer's Disease: A Systematic Review and Meta-Analysis. *Neuropsychol Rev* **32**, 193-227, doi:10.1007/s11065-021-09495-3 (2022).

28 Semmelmann, K. & Weigelt, S. Online webcam-based eye tracking in cognitive science: A first look. *Behav Res Methods* **50**, 451-465, doi:10.3758/s13428-017-0913-7 (2018).

29 Cuve, H. C., Stojanov, J., Roberts-Gaal, X., Catmur, C. & Bird, G. Validation of Gazepoint low-cost eye-tracking and psychophysiology bundle. *Behav Res Methods* **54**, 1027-1049, doi:10.3758/s13428-021-01654-x (2022).

30 Kaufmann, B. C. *et al.* Eyetracking during free visual exploration detects

ARTICLE IN PRESS

ARTICLE IN PRESS

neglect more reliably than paper-pencil tests. *Cortex* **129**, 223-235, doi:10.1016/j.cortex.2020.04.021 (2020).

31 Dintica, C. S. *et al.* The relation of poor mastication with cognition and dementia risk: a population-based longitudinal study. *Aging* **12**, 8536-8548, doi:10.18632/aging.103156 (2020).

32 Sta Maria, M. T., Hasegawa, Y., Khaing, A. M. M., Salazar, S. & Ono, T. The relationships between mastication and cognitive function: A systematic review and meta-analysis. *Jpn Dent Sci Rev* **59**, 375-388, doi:10.1016/j.jdsr.2023.10.001 (2023).

33 Mouelhi, Y., Jouve, E., Castelli, C. & Gentile, S. How is the minimal clinically important difference established in health-related quality of life instruments? Review of anchors and methods. *Health and Quality of Life Outcomes* **18**, 136, doi:10.1186/s12955-020-01344-w (2020).

34 *Instructions for Use for Controlled Medical Device*, <https://www.pmda.go.jp/PmdaSearch/kikiDetail/ResultDataSetPDF/113053_30500BZX00235000_A_01_04> (

35 Saito, T. & Rehmsmeier, M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PloS one* **10**, e0118432, doi:10.1371/journal.pone.0118432 (2015).

36 Mao, K. *et al.* Accuracy of near-infrared reflection in detecting proximal caries: a systematic review and meta-analysis. *J Dent* **161**, 105949, doi:10.1016/j.jdent.2025.105949 (2025).

37 Sasaki, Y. *et al.* Association between severity of malocclusion and parameters of oral functions in permanent dentition with various malocclusion: case-control study. *Clinical and Investigative Orthodontics*, 1-10, doi:10.1080/27705781.2023.2267847 (2023).

38 Kim, S. *et al.* Effect of chewing hard material on boosting brain antioxidant levels and enhancing cognitive function. *Front Syst Neurosci* **18**, 1489919, doi:10.3389/fnsys.2024.1489919 (2024).

39 Kulchutisin, P., Sowithayasakul, T., Pumklin, J. & Piyapattamin, T. Electromyographic Evaluations of Masticatory Muscle Activity between Patients with Skeletal Class I and III Relationships. *European journal of dentistry* **17**, 910-916, doi:10.1055/s-0042-1758064 (2023).

40 Sessle, B. J. Mechanisms of oral somatosensory and motor functions and their clinical correlates. *Journal of oral rehabilitation* **33**, 243-261, doi:10.1111/j.1365-2842.2006.01623.x (2006).

ARTICLE IN PRESS

ARTICLE IN PRESS

41 Avivi-Arber, L. & Sessle, B. J. Jaw sensorimotor control in healthy adults and effects of ageing. *Journal of oral rehabilitation* **45**, 50-80, doi:https://doi.org/10.1111/joor.12554 (2018).

42 Avivi-Arber, L. *et al.* Widespread Volumetric Brain Changes following Tooth Loss in Female Mice. *Front Neuroanat* **10**, 121, doi:10.3389/fnana.2016.00121 (2016).

43 Lin, C. S., Wu, S. Y., Wu, C. Y. & Ko, H. W. Gray Matter Volume and Resting-State Functional Connectivity of the Motor Cortex-Cerebellum Network Reflect the Individual Variation in Masticatory Performance in Healthy Elderly People. *Front Aging Neurosci* **7**, 247, doi:10.3389/fnagi.2015.00247 (2015).

44 Kubo, K. Y. *et al.* Tooth loss early in life suppresses neurogenesis and synaptophysin expression in the hippocampus and impairs learning in mice. *Archives of oral biology* **74**, 21-27, doi:10.1016/j.archoralbio.2016.11.005 (2017).

45 Kesner, R. P., Farnsworth, G. & Kametani, H. Role of parietal cortex and hippocampus in representing spatial information. *Cereb Cortex* **1**, 367-373, doi:10.1093/cercor/1.5.367 (1991).

46 Nitz, D. A. Tracking route progression in the posterior parietal cortex. *Neuron* **49**, 747-756, doi:10.1016/j.neuron.2006.01.037 (2006).

47 Baker, A. E., Galván, A. & Fuligni, A. J. The connecting brain in context: How adolescent plasticity supports learning and development. *Dev Cogn Neurosci* **71**, 101486, doi:10.1016/j.dcn.2024.101486 (2025).

48 Larsen, B. & Luna, B. Adolescence as a neurobiological critical period for the development of higher-order cognition. *Neurosci Biobehav Rev* **94**, 179-195, doi:10.1016/j.neubiorev.2018.09.005 (2018).

49 Poletti, B. *et al.* An eye-tracking controlled neuropsychological battery for cognitive assessment in neurological diseases. *Neurol Sci* **38**, 595-603, doi:10.1007/s10072-016-2807-3 (2017).

50 Athanasiou, A. E., Melsen, B., Mavreas, D. & Kimmel, F. P. Stomatognathic function of patients who seek orthognathic surgery to correct dentofacial deformities. *The International journal of adult orthodontics and orthognathic surgery* **4**, 239-254 (1989).

51 Ingervall, B. & Helkimo, E. Masticatory muscle force and facial morphology in man. *Archives of oral biology* **23**, 203-206, doi:10.1016/0003-9969(78)90217-0 (1978).

52 Chujo, M., Sugawara, J., Tomoyose, Y., Kawamura, H. & Mitani, H. Effects of

ARTICLE IN PRESS

ARTICLE IN PRESS

Functional Training with Chewing Gum after Surgical Orthodontic Treatment on Masticatory System in Jaw Deformity Patients. *The Japanese Journal of Jaw Deformities* **14**, 170-179, doi:10.5927/jjjd1991.14.170 (2004).

ARTICLE IN PRESS

ARTICLE IN PRESS

Figure legends

Figure 1. Chewing-induced BBF in NORM and MP.

(A) Representative temporal change of BBF during measurement. We set two tasks, calculation task and chewing task, with sufficient interval rest period to permit oxy-Hb concentrations to return to baseline levels.

(B) Comparison of oxidized hemoglobin (Oxy-Hb) values in the bilateral inferior frontal gyrus during chewing task in participants with normal occlusion (NORM) and patient with mandibular prognathism (MP). The figure shows percent induction of BBF in chewing task normalized by calculation task (%). Box plots represent median (horizontal line), mean (□), 25th and 75th percentiles (boxes), and minimum and maximum values (whiskers). Right and left inferior frontal gyrus in each group are presented. NORM values (n=17) are shown as open boxes, and MP values (n=44) are shown as gray boxes. * $p < 0.05$ between groups (Mann–Whitney U test).

Figure 2. Comparison of cognitive function scores between HC and MP.

Cognitive function scores measured by Mirudake are shown. Box plots represent median (horizontal line), mean (□), 25th and 75th percentiles (boxes), and minimum and maximum values (whiskers). Global cognitive performance score (A), memory (B), executive function (C), attention (D), visuospatial cognition (E), language (F), and orientation (G) are shown, respectively. HC values (n=59) are shown as open boxes, and MP values (n=44) are shown as gray boxes. NS: no statistically significant difference. * $p < 0.05$ between groups (Mann–Whitney U test).

ARTICLE IN PRESS

Figure 3. Heatmap visualization of the correlation among cognitive function scores.

The degree of correlation between the scores in Spearman (upper panel) and Pearson with bootstrap resampling (1000 iterations, lower panel) are shown. Heatmap color coding indicates correlation strength: 0.0-0.2 (weak correlation, white), 0.2-0.4 (weak to moderate correlation, blue), 0.4-0.7 (moderate to strong correlation, green), and 0.7-1.0 (strong correlation, red). NS: no statistically significant correlation.

Figure 4. ROC curve analysis to evaluate the discriminative ability of BBF measurements for cognitive status classification.

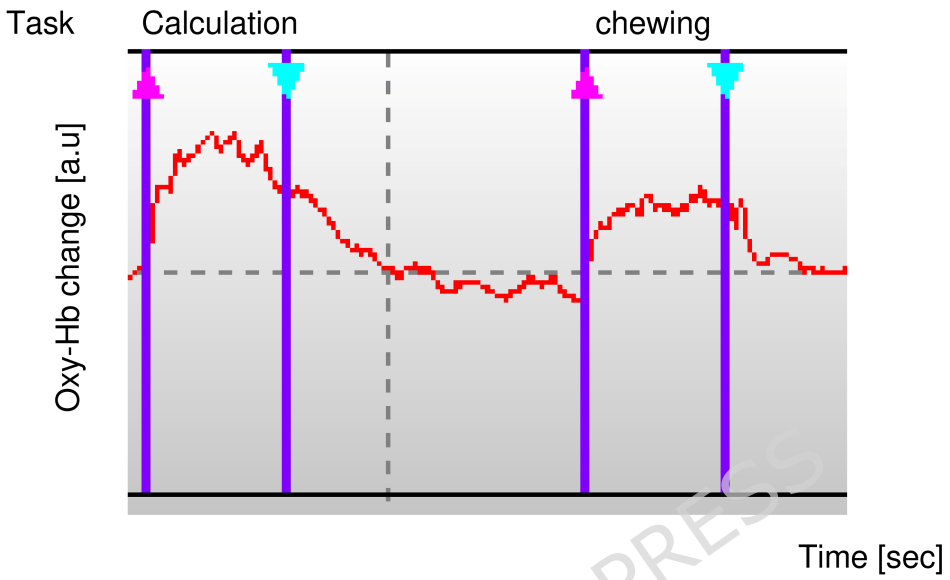
ARTICLE IN PRESS

BBF data from both left and right sides were pooled, resulting in a total sample size of 88 measurements (n = 88). The threshold was set as the Mirudake global cognitive performance score 50, which corresponding to the lower limit of the normal range in MMSE (i.e., MMSE ≥ 26), representing the boundary between normal cognitive function and cognitive impairment. This classification gave 8 values below threshold and 80 values above threshold. ROC curve suggesting that BBF measurements may have potential utility as an adjunct measure in cognitive screening. AUC = 0.657 (95% CI: 0.431-0.865, calculated with bootstrap resampling of 1000 iterations). Optimal threshold determined by Youden's index.

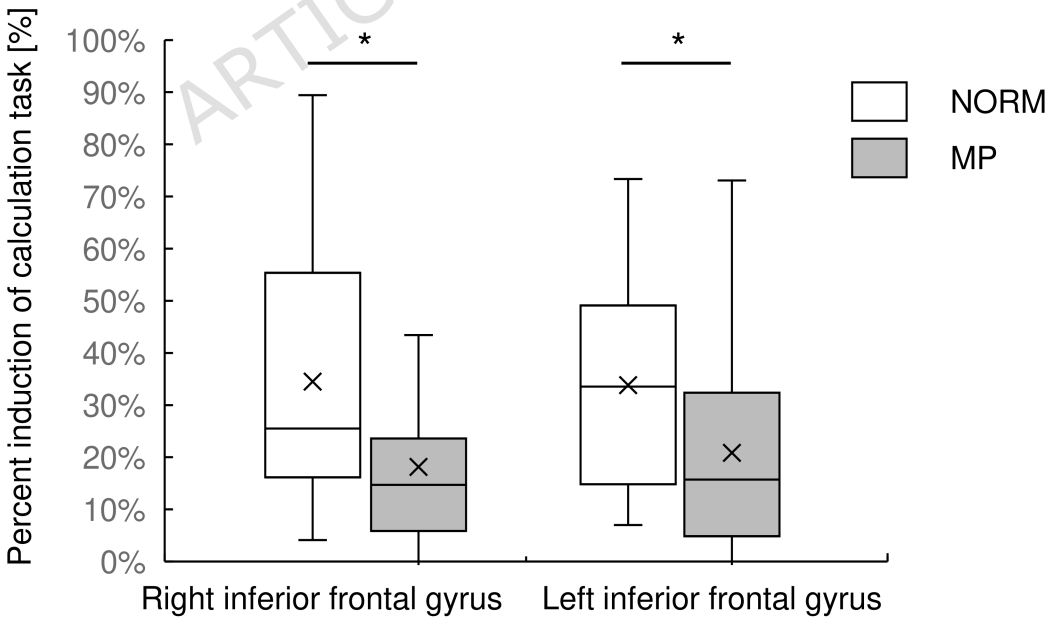
ARTICLE IN PRESS

ARTICLE IN PRESS

A

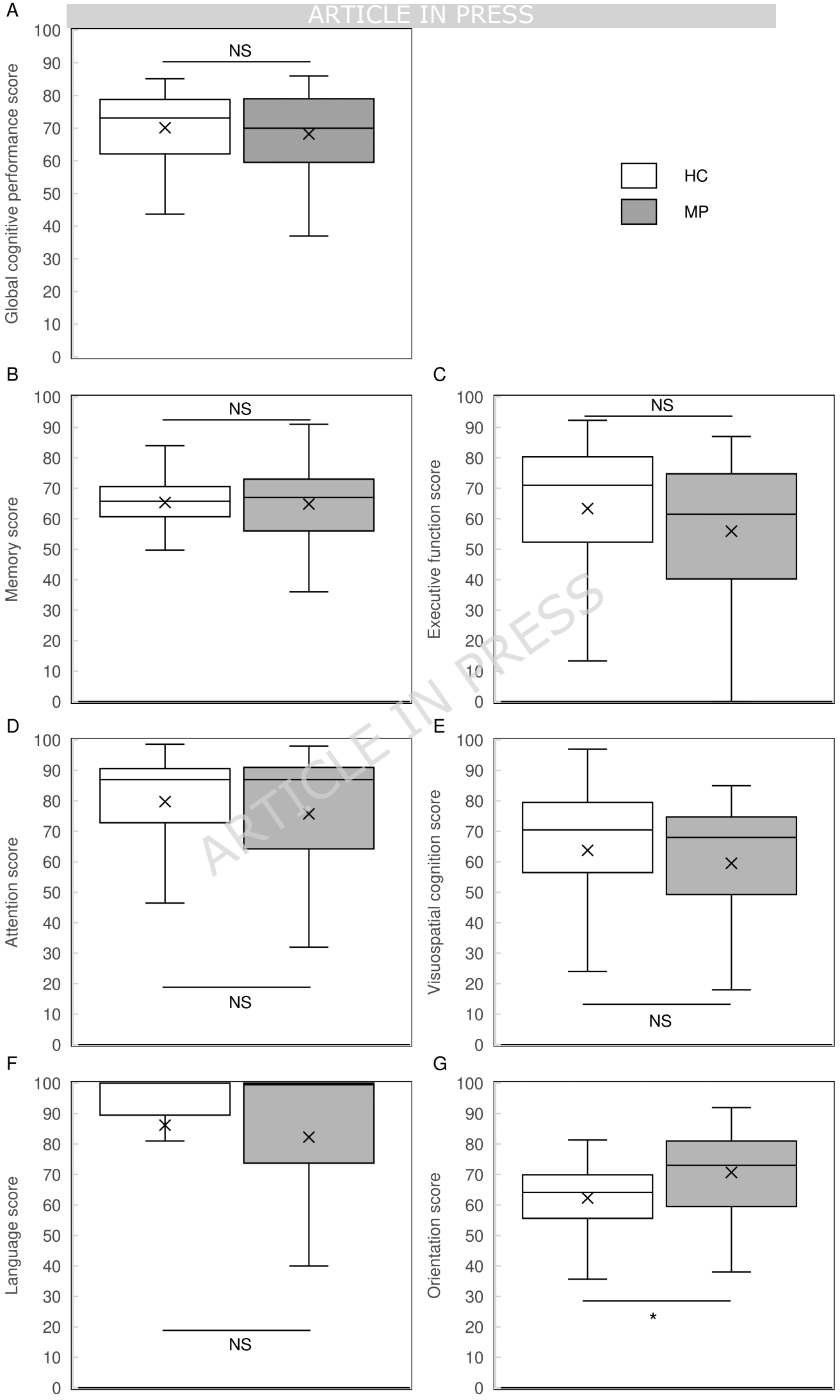


B



ARTICLE IN PRESS

ARTICLE IN PRESS



ARTICLE IN PRESS

ARTICLE IN PRESS

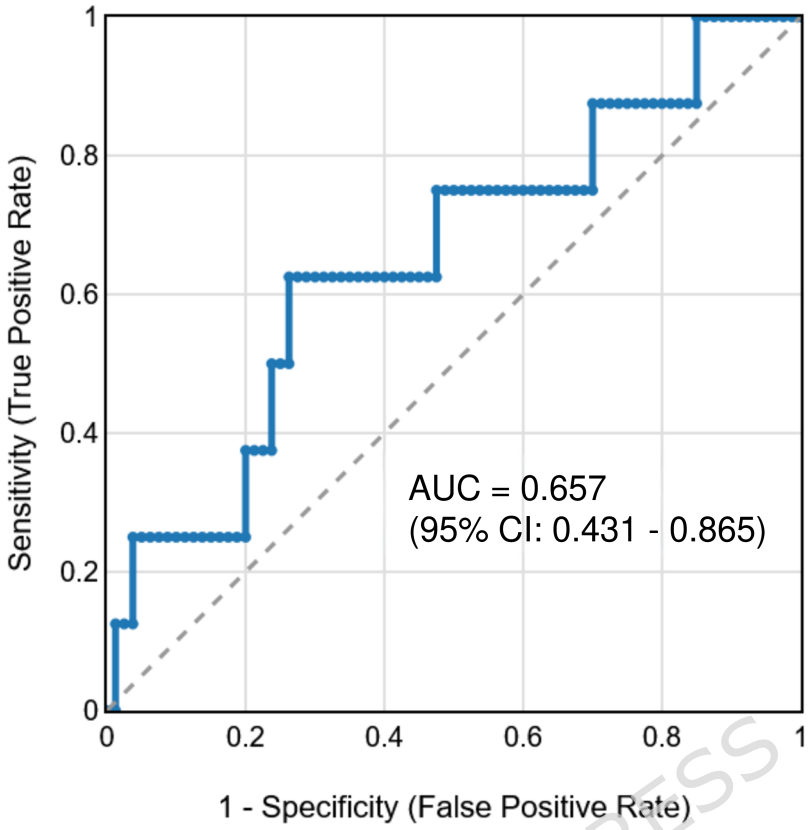
Spearman	Global cognitive performance score	Memory	Executive function	Attention	Visuospatial cognition	Language	Orientation
Global cognitive performance score	1.00	0.74	0.62	0.52	0.40	0.49	0.49
Memory		1.00	0.28	NS	0.30	0.55	NS
Executive function			1.00	NS	NS	NS	0.39
Attention				1.00	NS	NS	NS
Visuospatial cognition					1.00	NS	NS
Language						1.00	NS
Orientation							1.00

Pearson / Bootstrap	Global cognitive performance score	Memory	Executive function	Attention	Visuospatial cognition	Language	Orientation
Global cognitive performance score	1.00	0.75	0.66	0.47	0.61	0.58	0.36
Memory		1.00	0.36	NS	0.53	0.46	NS
Executive function			1.00	NS	0.32	NS	0.36
Attention				1.00	NS	NS	NS
Visuospatial cognition					1.00	NS	NS
Language						1.00	NS
Orientation							1.00

0.0 – 0.2
0.2 – 0.4
0.4 – 0.7
0.7 – 1.0

ARTICLE IN PRESS

ARTICLE IN PRESS



ARTICLE IN PRESS

ARTICLE IN PRESS

Table 1: Equivalence test results summary

Parameter	HC Group (n=59)	MP Group (n=44)	Difference (90% CI)	TOST results
Descriptive Statistics				
Median (IQR)	73.0 (62.0-79.0)	68.0 (59.5-78.0)	5.0	-
Mean*	69.97	68.41	1.56 [-2.03, 5.15]	-
Range	37-86	44-86	-	-
Equivalence Testing				
Conservative margin (±2.2)	-	-	CI exceeds margin	p_1=0.032, p_2=0.377 **Not equivalent**
Standard margin (±5.5)	-	-	CI within margin	p_1=0.001, p_2=0.035 **Equivalent**

Mean values used for equivalence testing calculations only. p_1 = lower bound test, p_2 = upper bound test. Equivalence requires both p-values < 0.05.

ARTICLE IN PRESS

ARTICLE IN PRESS

Table 2. Correlation of BBF against cognitive performance scores.

	Right inferior frontal gyrus		Left inferior frontal gyrus	
Parameter	Spearman	Pearson with Bootstrap	Spearman	Pearson with Bootstrap
Global cognitive performance score	0.323	0.3717 [0.0982, 0.5929]	0.393	0.3793 [0.114, 0.592]
	p < 0.01	p < 0.05	< 0.01	< 0.05
Memory	0.328	0.3452 [0.0759, 0.5552]	0.4205	0.3949 [0.164, 0.598]
	p < 0.01	p < 0.05	< 0.01	< 0.01
Executive function	0.130	0.157 [-0.120, 0.401]	0.3378	0.3423 [0.072, 0.549]
	NS	NS	< 0.05	< 0.05
Attention	0.191	0.142 [-0.141, 0.355]	0.1158	0.0699 [-0.228, 0.324]
	NS	NS	NS	NS
Visuospatial cognition	0.332	0.3160 [0.0846, 0.5003]	0.1756	0.2195 [-0.054, 0.424]
	p < 0.01	p < 0.05	NS	NS
Language	0.289	0.2254 [-0.0209, 0.4319]	0.1197	0.1468 [-0.073, 0.387]
	p < 0.05	NS	NS	NS
Orientation	-0.009	0.103 [-0.178, 0.366]	0.136	0.2192 [-0.110, 0.481]
	NS	NS	NS	NS

Correlation coefficient between BBF values and seven cognitive domains in Spearman analysis or Pearson with Bootstrap resampling with

ARTICLE IN PRESS

ARTICLE IN PRESS

1000 iterations are shown. 95% Confidence Intervals (CI) are also shown in Pearson with Bootstrap. NS: not statistically significance.

ARTICLE IN PRESS

ARTICLE IN PRESS

Table 3. Performance at Selected Thresholds

BBF Threshold	Sensitivity (%)	Specificity (%)	Accuracy (%)	Youden's Index
0.068174*	62.5	73.8	72.7	0.363
0.073537	62.5	72.5	71.6	0.350
0.077701	62.5	71.3	70.5	0.337
0.078867	62.5	70.0	69.3	0.325
0.079338	62.5	68.8	68.2	0.313
0.079649	62.5	67.5	67.0	0.300

Diagnostic performance at key threshold values. Optimal threshold marked with asterisk (*).

* Optimal threshold by maximum Youden's Index.

ARTICLE IN PRESS

ARTICLE IN PRESS

Table 4. Diagnostic Performance at Optimal Threshold

Metric	Value	95% CI
Sensitivity	62.50%	29.0-96.0%
Specificity	73.80%	64.1-83.4%
Positive Predictive Value	19.20%	4.1-34.4%
Negative Predictive Value	95.20%	89.8-100.0%
Accuracy	72.70%	63.4-82.0%
Area Under Curve	0.657	0.431-0.865

Performance characteristics at optimal threshold (0.068174) determined by Youden's index.

ARTICLE IN PRESS